VQE on two qubit Heisenberg model silently fails and outputs an energy lo

https://github.com/Qiskit/qiskit-aer/issues/1595

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VQE on two qubit Heisenberg model silently fails and outputs an energy lower than the ground state energy #1595

New issue

Closed

tomohiro-soejima opened on Sep 6, 2022

This is the same as Qiskit/qiskit#1587, but I realized `qiskit-aer` might be the right place to post it instead of `qiskit`. Please feel free to close either of these.

Informations

- **Qiskit version:**

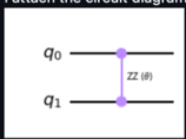
```
{'qiskit-terra': '0.21.1',
'qiskit-aer': '0.10.4',
'qiskit-ignis': None,
'qiskit-ibmq-provider': '0.19.2',
'qiskit': '0.37.1',
'qiskit-nature': None,
'qiskit-finance': None,
'qiskit-optimization': None,
'qiskit-machine-learning': None}
```

- **Python version:** `3.8.13`
- **Operating system:** Ubuntu 20:04 running on WSL2 on Windows 10

What is the current behavior?

When running VQE for the two spin Heisenberg model with the Hamiltonian `H = -(X^X) - (Y^Y) - (Z^Z)`, whose ground state energy is `-1`. However, when I use a custom ansatz and run VQE with `density_matrix` backend, VQE reports an energy of `-3`, which is lower than the ground state energy.

I attach the circuit diagram of the ansatz below:



Note that this is a rather pathological ansatz, in that the final state is independent of the parameter `theta`. Perhaps that is a cause of the problem?

Steps to reproduce the problem

Here is the MWE to reproduce the problem:

```
from inspect import Parameter
from qiskit import Aer
from qiskit.utils import QuantumInstance, algorithm_globals
from qiskit.algorithms import VQE, NumPyMinimumEigensolver
from qiskit.algorithms.optimizers import SPSA
from qiskit.opflow import I, X, Y, Z
from qiskit import QuantumCircuit
from qiskit.circuit import Parameter

def run_VQE(backend_str, operator, ansatz, ref_value):
    backend = Aer.get_backend(backend_str)
    seed = 175
    algorithm_globals.random_seed = seed
    qi = QuantumInstance(backend=backend, seed_simulator=seed)

    iterations = 10

    spsa = SPSA(maxiter=iterations)
    vqe = VQE(ansatz, optimizer=spsa, quantum_instance=qi)
    result1 = vqe.compute_minimum_eigenvalue(operator=operator)
    print(f'VQE on {backend_str}: {result1.eigenvalue.real:.5f}')
    print(f'Delta from reference energy value is {(result1.eigenvalue.real - ref_value):.5f}')

operator = -(X^X)  -(Y^Y) -(Z^Z)
npme = NumPyMinimumEigensolver()
```

Assignees
No one assigned

Labels
bug

Type
No type

Projects
No projects

Milestone
No milestone

Relationships
None yet

Development
No branches or pull requests

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```
# computes the reference eigenvalue from exact diagonalization
result = npme.compute_minimum_eigenvalue(operator=operator)
ref_value = result.eigenvalue.real
print(f'Reference value: {ref_value:.5f}')
qc = QuantumCircuit(2)
theta = Parameter(r"$\theta$")
qc.rzz(theta, 0, 1)
run_VQE("aer_simulator_density_matrix", operator, qc, ref_value)

# For comparison, we can run state vector simulation
# run_VQE("statevector_simulator", operator, qc, ref_value)
```

This prints out the following:

```
VQE on aer_simulator_density_matrix: -3.00000
Delta from reference energy value is -2.00000
```

What is the expected behavior?

VQE finds the energy of `-1`, which is the energy of the state `00`.

Suggested solutions

If we uncomment the final line of the MWE (`statevector_simulator` computation) and run it, it prints the following result and runtime warning messages:

```
VQE on statevector_simulator: nan
Delta from reference energy value is nan
/home/tomohiro/anaconda3/envs/large-vqe/lib/python3.8/site-packages/qiskit/algorithms/optimizers/spsa.py:345:
  a = target_magnitude / avg_magnitudes
/home/tomohiro/anaconda3/envs/large-vqe/lib/python3.8/site-packages/qiskit/algorithms/optimizers/spsa.py:565:
  update = update * next(eta)
```

This seems to signal there is a divide by zero error lurking around. Perhaps what is happening with the `density_matrix` calculation is some kind of silent divide by zero error?

tomohiro-soejima added bug on Sep 6, 2022

hhorii on May 16, 2023 Collaborator ···

I'm sorry that my response is very lazy.

With `statevector_simulator` or `BasicAer.get_backend("qasm_simulator")`, the above program does not work correctly with the current qiskit (though behavior of `aer_simulator_density_matrix` looks unexpected). In addition, operator flow is now deprecated. Please create a new issue if we encounter the same issue with the latest API.

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hhorii closed this as completed on May 16, 2023

tomohiro-soejima on May 17, 2023 Author ···

Thank you for your response!

jakelishman mentioned this on Aug 4, 2023

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Qiskit/qiskit-metapackage#1587

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